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INSTITUTE OF ENGINEERING
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**A Major Project Proposal
On
WIFI CSI HUMAN ACTIVITY RECOGNITION USING DEEP NEURAL
NETWORK**

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ABSTRACT

Human activity recognition has become increasingly important for applications such as smart homes, elderly care, and security monitoring, but conventional methods using cameras and wearable devices face challenges of privacy concerns, poor performance in low light, and the need for users to constantly wear sensors. This project presents a human activity recognition system that uses WiFi Channel State Information (CSI) to detect and classify human activities without any cameras or body-worn devices. Using three ESP32-S3 modules as the central CSI receivers, the system captures fine-grained wireless channel data from a WiFi router that acts as the WiFi signals transmitter. As a person moves within the area, their body alters the WiFi signals, producing distinct patterns in the CSI data that correspond to different activities such as walking, sitting, standing, lying down, falling. The collected CSI data is cleaned through filtering and phase correction, segmented into time windows, and then classified using deep learning models including a Convolutional Neural Network (CNN) and a CNN-LSTM hybrid. A labeled dataset is collected from multiple subjects in a controlled indoor environment to train and evaluate the system. This solution provides a low-cost, privacy-preserving, and contactless alternative to camera and wearable based systems, making indoor activity monitoring more practical and accessible using widely available WiFi hardware.

Keywords: Channel State Information (CSI), Long short-term memory (LSTM), Human Activity Recognition (HAR), Deep Learning, Convolutional Neural Network (CNN)

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
CSI	Channel State Information
DC	Direct Current
ESP-IDF	Espressif IoT Development Framework
FMCW	Frequency Modulated Continuous Wave
GPU	Graphics Processing Unit
HAR	Human Activity Recognition
IoT	Internet of Things
LSTM	Long Short-Term Memory
mmWave	Millimeter Wave
RNN	Recurrent Neural Network
RSS	Received Signal Strength
RSSI	Received Signal Strength Indicator
USB	Universal Serial Bus
WiFi	Wireless Fidelity

1. INTRODUCTION

1.1. Background

Wireless networks have become an inseparable part of modern indoor environments, with WiFi signals now present in nearly every home, office, hospital, and public building. While these signals are conventionally used only for data communication, the same radio waves continuously interact with the physical surroundings reflecting, scattering, and bending around walls, furniture, and human bodies. This incidental interaction carries rich information about the environment, and in recent years researchers have begun to exploit it to sense human presence and activity without any dedicated sensors. This emerging field, known as WiFi sensing, opens the possibility of turning ordinary wireless infrastructure into an intelligent sensing system.

Human Activity Recognition (HAR), the task of automatically identifying actions such as walking, sitting, standing, or falling, has wide-ranging applications in smart homes, elderly care, healthcare monitoring, and security. Traditionally, HAR has relied on cameras or wearable sensors. However, these conventional approaches face notable limitations. Camera-based systems raise serious privacy concerns, fail in poor lighting, and cannot see through obstacles, while wearable devices require users to continuously carry or wear them, which is often impractical, especially for elderly individuals who may forget or refuse to use them. These drawbacks have motivated the search for a contactless, privacy-preserving, and always-available alternative.

WiFi Channel State Information (CSI) has emerged as a promising solution to this problem. Unlike the simple signal strength measurement (RSSI) used in older systems, CSI provides fine-grained, physical-layer information describing how each frequency component of a WiFi signal is affected by the channel between a transmitter and a receiver. When a person moves within the coverage area, their body perturbs the multipath propagation of the signal, producing distinct and repeatable patterns in the CSI. By analyzing these patterns, it is possible to recognize different human activities in a completely device-free manner, without cameras and without requiring the person to wear anything.

Despite the strong potential of CSI-based sensing, most existing research depends on specialized and increasingly obsolete hardware such as the discontinued Intel 5300 network card, or on complex firmware modifications to laptop-grade wireless chipsets. Such hardware is difficult to source and expensive in markets like Nepal, where importing specialized com-

ponents introduces additional cost and delay. This limits the ability of students and local researchers to study and reproduce CSI-based systems. The recent availability of the low-cost ESP32 microcontroller, together with the open-source ESP32-CSI-Tool firmware, has made CSI data accessible on inexpensive and locally available hardware, removing a major barrier to entry and enabling practical experimentation within a modest budget.

This project focuses on the design and implementation of a WiFi CSI-based Human Activity Recognition system using low-cost and readily available hardware. The system uses three ESP32-S3 modules as CSI receivers and a router as the WiFi transmitter to capture channel information as a person performs various activities within an indoor space. The collected CSI data is processed to remove noise and hardware-induced distortions, segmented into time windows, and then classified into different activities using deep learning models. By building a contactless recognition system on affordable and accessible hardware, the proposed project aims to demonstrate a practical, privacy-preserving, and reproducible approach to human activity sensing suitable for local conditions and future research.

1.2. Problem Statement

Despite the increasing demand for indoor human activity monitoring in healthcare, elderly care, smart homes, and security systems, existing solutions still face several practical limitations that restrict their real world deployment, especially in developing countries like Nepal. Camera based systems raise serious privacy concerns and are highly dependent on lighting conditions, making them unsuitable for private or low visibility indoor environments. Wearable sensor-based systems, although effective, require continuous user compliance, regular charging, and long-term maintenance, which makes them impractical for elderly users, patients, and passive monitoring scenarios. On the other hand, radar based solutions such as mmWave and FMCW systems provide high accuracy but are too expensive for large-scale or household-level deployment in resource-constrained settings. WiFi CSI-based approaches have emerged as a promising alternative; however, most existing research relies on outdated or specialized hardware that is no longer easily accessible, and many systems are trained on controlled laboratory datasets that fail to generalize well to real-world indoor environments where layout, materials, and human behavior vary significantly. As a result, there is a clear gap between research prototypes and practical, deployable systems. Therefore, there is a need for a low-cost, privacy-preserving, and easily deployable human activity recognition system that utilizes modern and accessible hardware such as ESP32-S3 modules, leverages WiFi CSI signals, and applies deep learning techniques for robust performance in real-world conditions.

1.3. Objectives

1. To design and implement a multi-receiver CSI acquisition setup using three ESP32-S3 devices and a standard WiFi router, and to collect a labeled dataset of human activities in an indoor environment.
2. To develop a signal preprocessing pipeline that filters and corrects the raw CSI data collected from the ESP32-S3 modules.
3. To develop and evaluate deep learning models (CNN and CNN-LSTM) capable of learning the CSI patterns and classifying human activities such as walking, sitting, standing, lying down, and falling.

1.4. Scope

The scope of this WiFi CSI human activity recognition project covers the following areas:

- **Privacy-Preserving Monitoring:** Enables contactless detection of human activities such as walking, sitting, lying down and falling without using cameras or microphones, making it suitable for sensitive spaces like bedrooms, bathrooms, and elderly-care rooms where visual surveillance is unacceptable.
- **Low-Cost Accessibility:** Uses inexpensive and locally available ESP32-S3 modules with a commodity WiFi router instead of costly or discontinued specialized hardware, bringing CSI-based sensing within reach of students, researchers, and small-scale deployments in resource-limited settings like Nepal.
- **Device-Free Operation:** Detects activity by analyzing how the human body perturbs WiFi signals, requiring no wearable devices or active participation from the user, which is especially valuable for monitoring elderly or vulnerable individuals who cannot reliably carry or wear sensors.
- **Reproducible Research Foundation:** Delivers a locally collected dataset, trained deep learning models, and fully documented source code, establishing a reusable baseline that future student projects can extend toward applications such as fall detection, occupancy sensing, and smart-home integration.

1.5. Feasibility Study

- **Technical Feasibility:** The proposed system is technically feasible as it uses readily available hardware such as ESP32-S3 microcontrollers and a standard WiFi router for CSI data collection. WiFi CSI-based sensing is supported by existing open-source tools like ESP32-CSI-Tool, making data acquisition achievable without specialized hardware. The processing and classification tasks can be handled using a standard laptop with commonly used deep learning frameworks such as Python, PyTorch. Therefore, the required tools, hardware, and software are accessible and sufficient for successful implementation of the system within the project timeframe.
- **Economic Feasibility:** The system is economically feasible because it relies on low-cost and easily available components. ESP32-S3 modules and a standard WiFi router are affordable compared to alternatives such as radar-based sensing systems or specialized camera setups. Additionally, the project does not require expensive sensors, cloud services, or proprietary software licenses. Since most development tools used are free, the overall project cost remains within a student level budget, making it suitable for implementation in a resource constrained environment.
- **Operational Feasibility:** The system is operationally feasible as it is designed to be simple, non-intrusive, and easy to use in real indoor environments. It does not require users to wear devices or interact with the system during operation, making it practical for continuous monitoring scenarios. Once trained and deployed, the model can perform real-time activity recognition with minimal user involvement. The system is suitable for applications such as basic fall detection and indoor monitoring, and can be demonstrated effectively in a laboratory or home environment for testing and evaluation purposes.

2. LITERATURE REVIEW

- **Ding and Wang [1]** proposed a CSI-based human activity recognition system based on WiFi using a deep recurrent neural network (RNN). Their HARNN model classified activities such as walking, sitting, and falling with high accuracy. However, the work was mainly evaluated in controlled indoor environments and did not address multi-user scenarios.
- **Yousefi et al. [2]** presented a comprehensive review of WiFi CSI-based human activity recognition. The paper highlighted that CSI provides more detailed channel information than RSS-based systems, allowing accurate activity and gesture recognition. Key challenges identified include environmental interference, cross-domain adaptability, and real-time deployment.
- **Abdelnasser et al. [3]** developed WiGest, a gesture recognition framework using WiFi signal variations to detect hand gestures without wearables or cameras. The system analyzed variations in WiFi signal strength (RSSI) for gestures such as pushing and swiping. Performance was affected by variations in environmental noise and movement.
- **Gu et al. [4]** introduced WiGRUNT, a dual-attention deep learning model for WiFi CSI-based gesture recognition combining spatial and temporal attention mechanisms over a ResNet backbone. Their approach achieved improved cross-domain accuracy over traditional CNN and RNN models for complex gesture patterns.
- **Chen et al. [5]** proposed a lightweight deep learning framework using a gated recurrent unit (GRU) with an attention module for device-free HAR using reconstructed WiFi CSI. The system achieved robust classification while significantly reducing model size and computational cost, making it more suitable for resource-constrained embedded deployment.
- **Miao et al. [6]** reviewed recent advances in WiFi sensing for human activity recognition using deep learning, surveying methodologies, signal-collection tools, and public datasets. Key research gaps include dataset generalization, environmental adaptability, multi-person interference, and real-time processing.
- **Fard Moshiri et al. [7]** investigated deep learning for indoor HAR using WiFi CSI, demonstrating that CSI provides more stable features than RSS-based approaches. Their system recognized seven activities including walking, standing, and sitting but faced challenges with signal noise and model generalization.

3. METHODOLOGY

3.1. Basic Working

The WiFi router acts as the signal transmitter and continuously sends wireless packets over a fixed channel. These signals spread throughout the indoor environment and interact with everything in the room, including any person present in the sensing area.

Three ESP32-S3 modules act as the central receiving units and are placed at fixed positions around the area. Each module runs the ESP32-CSI-Tool firmware and captures the Channel State Information (CSI) from the received packets, which represents how the WiFi signal is affected as it travels through the space. When a person performs an activity such as walking, sitting, falling, lying down their body alters the signal in a unique way, producing distinct patterns in the captured CSI.

The CSI data collected by the three ESP32-S3 modules is sent to a host computer through a USB connection. Each CSI record is tagged with an arrival timestamp on the host computer. The host then aligns the three receiver streams into common fixed length time windows within a small tolerance. Because the three modules run on independent clocks, exact per packet correspondence is not assumed. Instead, window level alignment is used, and residual synchronization error is treated as a known limitation

The collected raw CSI data is then preprocessed to prepare it for recognition. This includes removing noise and unwanted variations using filtering techniques, correcting signal distortions, and dividing the continuous data into fixed time windows. Each window is normalized so that it can be fed into the recognition model.

Based on the patterns in each time window, a trained deep learning model classifies the data into one of the predefined activity classes such as walking, sitting, standing, lying down, falling. Two models, a Convolutional Neural Network (CNN) and a CNN-LSTM hybrid, are used and compared to identify which gives better recognition accuracy.

Finally, the recognized activity is given as the system output and stored for evaluation. The predicted activities are compared with the actual activities performed to measure the accuracy and performance of the system. This methodology enables contactless and privacy-preserving human activity recognition using only low-cost WiFi hardware, without the need for cameras or wearable devices.

3.2. Block Diagram

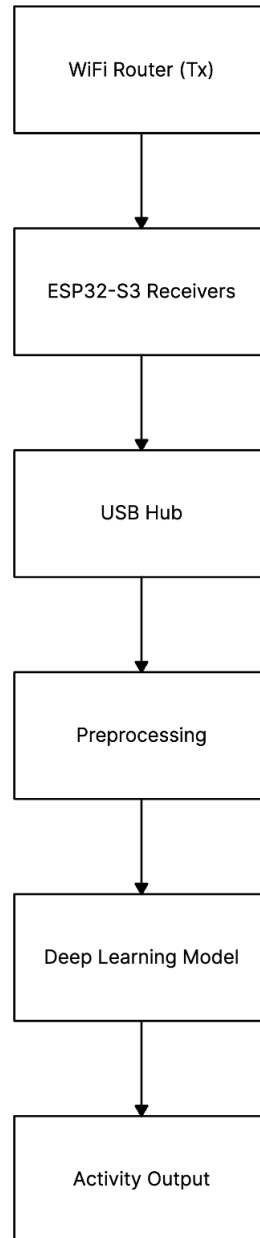


Fig 3.2.1: Working Block Diagram

3.3. Flowchart

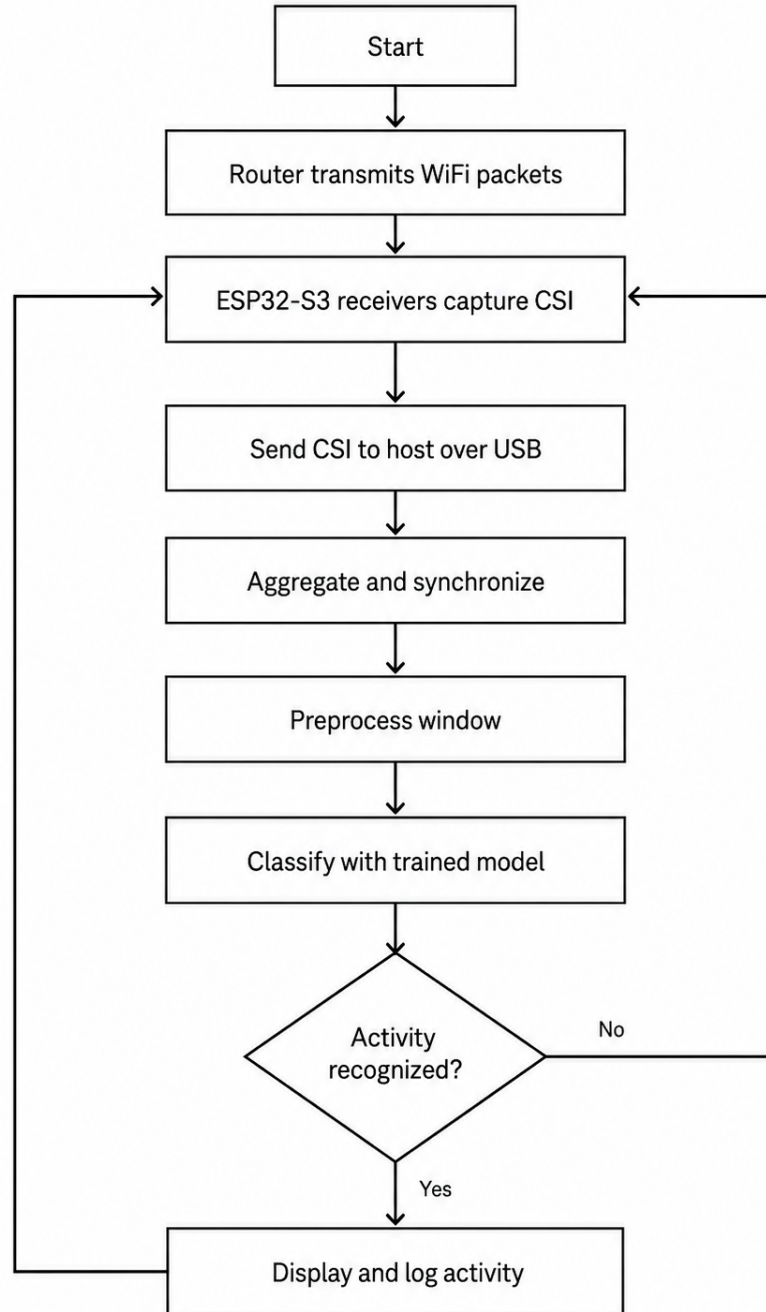


Fig 3.3: Flowchart of the Complete Workflow

4. IMPLEMENTATION DETAILS

4.1. Hardware Requirements

4.1.1. ESP32-S3 Microcontroller

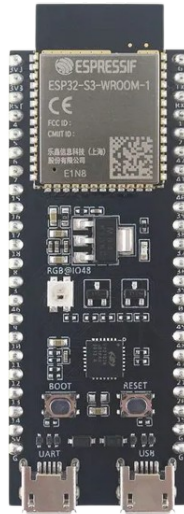


Fig 4.1.1: ESP32-S3 Microcontroller

The ESP32-S3 captures WiFi Channel State Information (CSI) from the packets transmitted by the router. Each module runs the ESP32-CSI-Tool firmware, which extracts the channel measurements from the WiFi driver and outputs them as readable CSI records. These records are streamed to the host computer over the USB-serial interface for preprocessing and classification. CSI is obtained directly from the standard WiFi radio already on the chip, no additional radio hardware is required, keeping the system small, low-cost, and power-efficient while preserving privacy.

4.1.2. Router



Fig 4.1.2: Router

The WiFi router acts as the signal transmitter in a human activity recognition system. It continuously emits WiFi signals, which are received by the ESP32-S3. Human movements in the environment cause small changes in these signals, which are captured as Channel State Information (CSI). The router requires no modification and provides a stable, low-cost, and privacy-preserving way to enable wireless sensing without cameras or wearable devices.

4.1.3. Battery



Fig 4.1.3: Battery

The battery provides portable power to the ESP32-S3 and ensures continuous operation of the human activity recognition system. It supplies stable DC power, allowing the device to function without being connected to a constant power source. This makes the system mobile, low-cost, and suitable for real-world deployment.

4.1.4. Battery Holder



Fig 4.1.4: Battery Holder

The battery holder securely holds the battery and provides a stable connection to power the ESP32-S3. It ensures continuous and reliable power supply, allowing the setup to operate without external wiring. This makes the system more portable, organized, and suitable for real-world deployment and testing.

4.1.5. Tripod



Fig 4.1.5: Tripod

The tripod is used to hold the ESP32-S3 setup at a fixed position and height during data collection. It ensures stability and consistent placement of devices, which improves the quality and reliability of the captured signal data. This helps maintain uniform experimental conditions for the human activity recognition system.

4.2. Software Requirements

4.2.1. ESP-IDF and ESP32-CSI-Tool

The ESP32-S3 modules are programmed using the Espressif IoT Development Framework (ESP-IDF), the official development toolchain for ESP32 devices. ESP-IDF was chosen over the Arduino framework because CSI extraction requires low-level access to the WiFi driver, which ESP-IDF exposes directly. On top of this framework, the open-source ESP32-CSI-Tool firmware is flashed onto each module. This firmware modifies the standard WiFi behavior so that, instead of discarding the channel measurements after each received packet, the module extracts the Channel State Information and outputs it as readable text. Each ESP32-S3 is configured in station mode, connected to the router on a fixed channel, and set to forward its CSI records to the host computer over the USB-serial interface.

4.2.2. Python

Python is used as the primary programming language for all host-side work, including data collection, preprocessing, model development, and evaluation. It is chosen for its simplicity, large ecosystem of scientific libraries, and strong support for machine learning. The data collection scripts use the pyserial library to read CSI streams from all three modules at the same time, while NumPy, SciPy, and Pandas are used to clean, filter, and organize the collected data. SciPy's signal-processing functions are used specifically to apply the Hampel and low-pass filters during preprocessing.

4.2.3. PyTorch

The deep learning models are built and trained using PyTorch, an open-source machine learning framework widely used in both research and industry. PyTorch is selected for its flexibility, clear syntax, and good support for building custom network architectures such as the CNN and CNN-LSTM models used in this project. It also supports GPU acceleration, which speeds up training when a compatible graphics card is available. PyTorch handles the full training process, including loading the prepared dataset, updating the model during training, and saving the best-performing model for later evaluation.

4.2.4. Visualization Tools

Matplotlib and Seaborn are used to visualize data and results throughout the project. Matplotlib is used to plot raw and processed CSI signals, which helps verify that the preprocessing steps are working correctly and that different activities produce visibly different patterns. Seaborn is used to create clearer statistical plots such as confusion matrices and per-class performance charts, which make the evaluation results easier to interpret and present.

4.2.5. Git and GitHub

Git is used as the version control system to track changes in the project's code, and GitHub is used as the online repository to store the code and allow all four team members to collaborate. This ensures that the work is backed up, that changes made by different members can be merged safely, and that a complete history of the project's development is maintained.

4.3. Cost Estimation

S.N.	Component	Cost (NRs)
1	ESP32 S3-N16R8	6600
2	Tripod	1050
3	Li-ion Battery	1200
4	Router	3000
5	Data Cables	3000
6	Battery Holder	600
7	Miscellaneous	3000
Total		18450

Table 4.3: Cost Estimation

4.4. Timeline

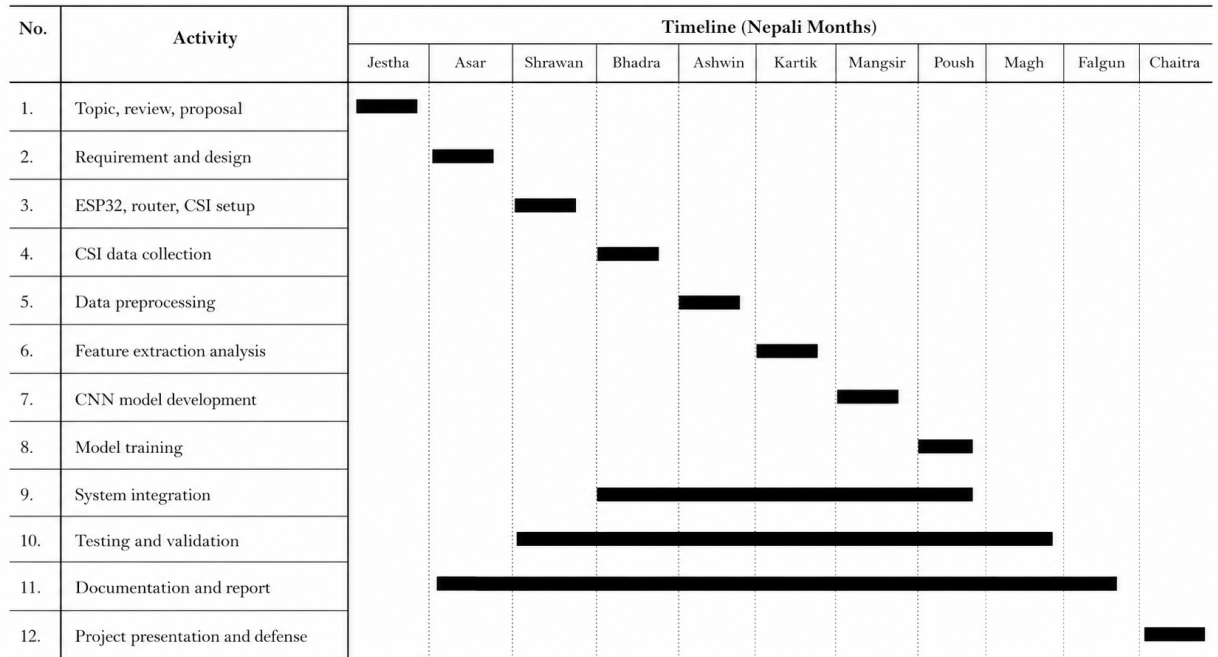


Fig 4.4: Gantt Chart

5. EXPECTED OUTCOME

The proposed system is expected to detect and classify common human activities such as walking, sitting, standing, lying down and falling using only WiFi signals, without any cameras or wearable devices. This will show that human activity recognition can be done in a contactless and privacy-preserving way.

By using three low-cost ESP32-S3 modules and a commodity WiFi router, the system is expected to prove that reliable activity recognition is possible with inexpensive and locally available hardware, making it well suited for students and small-scale use in settings like Nepal.

The system is aimed to achieve a classification accuracy of at least $\geq 85\%$ across the defined activities, based on a CSI dataset collected in a local environment. The comparison between the CNN and CNN-LSTM models will show which performs better, and the use of three receivers is expected to improve accuracy over a single receiver.

The project is expected to deliver a complete and reproducible set of outputs, including the dataset, trained models, and documented source code. These can be extended in the future toward practical applications such as fall detection, occupancy sensing, and smart-home monitoring. Overall, the project aims to deliver a low-cost, privacy-preserving, and reproducible activity recognition system that provides a strong foundation for future research.

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